

Model Lab

Every prior, same slate — 2026-09-24

SERVED PRIOR

Core (KNN+SVR)

PRIORS COMPARED

10

HELD-OUT PA

37,564

Held-Out Scores

Fitted on plate appearances before 2026-08-17, scored on the 37,564 that came after (actual HR rate 2.93%). No outcome in the test window was visible during fitting.

Metrics are per plate appearance; the probabilities lower down are per game.

Read AUC first — the slate ranks hitters and takes the top six, so ordering is what matters. Brier and log loss are dominated by the ~3% base rate and barely separate the priors.

Prior	AUC	Top-decile lift	Brier	Log loss	Cal. RMSE
Flat league League HR rate as the prior for every hitter.	0.5736	1.53×	0.02840	0.13151	0.00303
KNN comps Prior is the pooled rate of the nearest comparables.	0.5886	1.46×	0.02840	0.13118	0.00495
SVR Support-vector regression on the same feature matrix.	0.5833	1.39×	0.02850	0.13263	0.00853
Random forest Forest regression; a learned similarity metric.	0.5816	1.44×	0.02850	0.13236	0.00781
Core (KNN+SVR) (served) Blend of KNN and SVR, no forest.	0.5886	1.46×	0.02840	0.13118	0.00495
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5884	1.44×	0.02841	0.13122	0.00507
Linear regression Ordinary least squares on the same features.	0.5854	1.37×	0.02849	0.13231	0.00775
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5863	1.49×	0.02853	0.13215	0.00803
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5888	1.44×	0.02848	0.13184	0.00784
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5844	1.50×	0.02855	0.13257	0.00868

Blend weights (forest variant): KNN 0.95, SVR 0.00, RF 0.05

Shrinkage K per prior: league 248, knn 248, svr 248, rf 248, core 248, ensemble 248, linear 248, logistic 248, nn 248, xgb 248 — estimated by method of moments on out-of-fold residuals, not hand-tuned.

Thin-Sample Subset

The same held-out window, restricted to hitters with fewer than 100 plate appearances before the cutoff (4,400 PA, actual HR rate 2.34%). For a regular with 600 PA the observed record swamps any prior, so the priors mostly separate here.

Prior	AUC	Top-decile lift	Brier	Log loss	Cal. RMSE
Flat league League HR rate as the prior for every hitter.	0.5113	1.26×	0.02292	0.11197	0.00982
KNN comps Prior is the pooled rate of the nearest comparables.	0.5519	0.97×	0.02310	0.11398	0.01675
SVR Support-vector regression on the same feature matrix.	0.5702	0.97×	0.02361	0.12065	0.02681
Random forest Forest regression; a learned similarity metric.	0.5460	0.49×	0.02373	0.12035	0.03060
Core (KNN+SVR) (served) Blend of KNN and SVR, no forest.	0.5519	0.97×	0.02310	0.11398	0.01675
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5525	0.97×	0.02312	0.11420	0.01702
Linear regression Ordinary least squares on the same features.	0.5503	0.58×	0.02377	0.12136	0.03001
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5481	0.58×	0.02399	0.12005	0.03349
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5663	0.87×	0.02364	0.11784	0.02913
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5735	0.68×	0.02406	0.12142	0.03531

Slate Agreement

Flat league · 4/6 shared 0/5 hit

Ben Rice
Miguel Vargas
Yordan Alvarez
Jeremy Peña
Ronald Acuña Jr.
Luis García Jr.

SVR · 6/6 shared 0/5 hit

Ben Rice
Yordan Alvarez
Miguel Vargas
Kyle Schwarber
Luis García Jr.
Matt Olson

Core (KNN+SVR) · 6/6 shared 0/5 hit

Ben Rice
Miguel Vargas
Yordan Alvarez
Kyle Schwarber
Luis García Jr.
Matt Olson

Linear regression · 6/6 shared 0/5 hit

Ben Rice
Yordan Alvarez
Miguel Vargas
Kyle Schwarber
Matt Olson
Luis García Jr.

Neural net (torch) · 6/6 shared 0/5 hit

KNN comps · 6/6 shared 0/5 hit

17.5% Ben Rice 19.9%
16.6% Miguel Vargas 19.5%
16.2% Yordan Alvarez 18.7%
15.7% Kyle Schwarber 17.0%
15.6% Luis García Jr. 17.0%
15.5% Matt Olson 16.9%

Random forest · 6/6 shared 0/5 hit

21.4% Ben Rice 20.4%
20.9% Yordan Alvarez 20.1%
19.6% Miguel Vargas 19.2%
18.6% Kyle Schwarber 18.1%
18.5% Luis García Jr. 17.7%
18.4% Matt Olson 17.6%

Forest blend · 6/6 shared 0/5 hit

19.9% Ben Rice 19.9%
19.5% Miguel Vargas 19.5%
18.7% Yordan Alvarez 18.8%
17.0% Kyle Schwarber 17.0%
17.0% Luis García Jr. 17.0%
16.9% Matt Olson 16.9%

Logistic regression · 6/6 shared 0/5 hit

20.1% Yordan Alvarez 21.3%
19.8% Ben Rice 20.4%
19.8% Miguel Vargas 20.3%
17.8% Kyle Schwarber 19.9%
17.1% Matt Olson 18.5%
17.1% Luis García Jr. 17.9%

Luis García Jr.

17.3%

Luis García Jr.

17.7%

Hitter by Hitter

[illegible]

Hitter	Flat league	KNN comps	SVR	Random forest	Core (KNN+SVR)	Forest blend	Linear regression	Logistic regression	Neural net (torch)
8. Kyle Stowers slate Miami Marlins · vs RHP · 21 HR / 511 PA	14.0%	16.8%	17.7%	16.7%	16.8%	16.8%	16.3%	17.2%	16.9%
9. Jeremy Peña Houston Astros · vs RHP · 20 HR / 461 PA	15.7%	16.1%	16.7%	16.5%	16.1%	16.1%	15.8%	16.0%	16.2%
10. Pete Crow-Armstrong Chicago Cubs · vs LHP · 45 HR / 712 PA	14.4%	16.0%	17.3%	16.8%	16.0%	16.1%	16.5%	17.7%	16.9%
11. Corey Seager Texas Rangers · vs RHP · 18 HR / 420 PA	12.9%	16.0%	15.7%	16.0%	16.0%	16.0%	15.9%	16.0%	16.2%
12. Munetaka Murakami Chicago White Sox · vs RHP · 33 HR / 535 PA	13.2%	15.8%	16.4%	16.4%	15.8%	15.8%	15.7%	15.7%	15.9%
13. Heriberto Hernández Miami Marlins · vs RHP · 24 HR / 489 PA	13.5%	15.7%	15.8%	15.9%	15.7%	15.7%	15.6%	15.7%	15.9%
14. Brandon Lowe Pittsburgh Pirates · vs RHP · 34 HR / 642 PA	13.8%	15.5%	16.0%	16.0%	15.5%	15.6%	15.9%	16.8%	15.9%
15. Leo Bernal St. Louis Cardinals · vs RHP · 4 HR / 86 PA	12.4%	15.5%	14.4%	15.6%	15.5%	15.5%	15.9%	14.1%	17.1%
16. Cal Raleigh Seattle Mariners · vs RHP · 23 HR / 516 PA	12.8%	15.5%	14.3%	14.7%	15.5%	15.4%	15.7%	16.7%	15.5%
17. Elly De La Cruz Cincinnati Reds · vs RHP · 29 HR / 618 PA	14.5%	15.5%	16.6%	15.8%	15.5%	15.5%	15.5%	15.9%	15.8%
18. Eugenio Suárez Cincinnati Reds · vs RHP · 25 HR / 486 PA	14.2%	15.4%	15.7%	15.5%	15.4%	15.4%	15.5%	15.7%	14.8%
19. Juan Soto New York Mets · vs LHP · 27 HR / 463 PA	12.5%	15.4%	16.6%	15.6%	15.4%	15.4%	15.6%	15.8%	16.0%
20. Oneil Cruz Pittsburgh Pirates · vs RHP · 20 HR / 410 PA	13.3%	15.3%	14.7%	14.8%	15.3%	15.3%	14.9%	14.8%	15.2%
21. Jackson Chourio Milwaukee Brewers · vs RHP · 24 HR / 561 PA	14.4%	15.2%	15.7%	15.9%	15.2%	15.2%	15.2%	15.4%	15.3%
22. Lazaro Montes	11.4%	15.1%	13.0%	14.2%	15.1%	15.1%	14.8%	14.4%	13.7%

Hitter	Flat league	KNN comps	SVR	Random forest	Core (KNN+SVR)	Forest blend	Linear regression	Logistic regression	Neural net (torch)
37. Mickey Moniak Colorado Rockies · vs RHP · 23 HR / 431 PA	12.5%	14.4%	15.4%	14.6%	14.4%	14.4%	14.1%	14.9%	14.6%
38. Drake Baldwin Atlanta Braves · vs RHP · 24 HR / 601 PA	13.0%	14.2%	14.5%	14.8%	14.2%	14.3%	14.5%	14.0%	14.3%
39. William Contreras Milwaukee Brewers · vs RHP · 18 HR / 628 PA	13.9%	14.2%	14.0%	13.8%	14.2%	14.2%	13.8%	13.9%	13.8%
40. Francisco Lindor New York Mets · vs LHP · 20 HR / 450 PA	13.3%	14.2%	14.0%	13.6%	14.2%	14.2%	14.1%	14.0%	14.0%

Per-game probabilities: each prior's per-PA rate run through this hitter's own plate appearances, park, weather, bullpen and times-through-the-order terms. Only the prior changes across the columns — every other term is held fixed, so a left-to-right spread is the prior disagreeing about the hitter, not the context.